FEU INSTITUTE OF TECHNOLOGY

COURSE CODE	COURSE TITLE	UNITS / TYPE
IT0013	NETWORKING 1	2 UNITS LEC/ 1 UNIT LAB
PREREQUISITE	INTRODUCTION TO COMPUTING	
CO-REQUISITE/S	NONE	
COURSE DESCRIPTION	The course introduces the architectures, models, protocols, and networking elements that connect users, devices, applications and data through the Internet and across modern computer networks - including IP addressing and Ethernet fundamentals. By the end of the course, students can build simple local area networks (LAN) that integrate IP addressing schemes, foundational network security, and perform basic configurations for routers and switches. Packet tracer will be used as the simulation program for all computer lab activities.	

INSTITUTION VISION STATEMENT

FEU Institute of Technology aims to be the global standard of innovation in education, research, and nation-building by creating the next generation of changemakers that will shape the future of the country and the world.

INSTITUTION MISSION STATEMENT

FEU Institute of Technology is dedicated to provide quality, relevant, innovative and industry-based education producing competent and principled professionals with greater sense of responsibility, social awareness and high competitiveness contributing significantly to the betterment of the society.

DEPARTMENT VISION STATEMENT

The Information Technology Department aims its program specializations to be a catalyst on the delivery of industry-based standards solutions and internationally recognized IT education.

DEPARTMENT MISSION STATEMENT

The Information Technology Department is committed to provide industry- based information technology solutions, international academic linkages, researches and IT certified professionals.

EDUCATIONAL PHILOSOPHY

FEU Tech's educational philosophy is rooted in transformative learning, emphasizing innovation, relevance, and social responsibility. We aim to cultivate an ecosystem of learning that shapes the next generation of leaders and innovators, driving positive change both locally and globally.

PROGRAM EDUCATIONAL OUTCOMES

The graduates of the Bachelor of Science in Information Technology program are:

1. engaged in further professional development and have interest in or aptitude for advanced studies or trainings in computing.
2. entrepreneurs or are employed in computing industries, organizing and managing team-based projects leading to successful and sustainable computing systems solutions.
3. responsible computing professionals actively participating in community groups that make a significant impact in addressing current and future societal challenges.

PROGRAM OUTCOMES

a. Apply knowledge of computing appropriate to the discipline.
b. Analyze a complex problem and identify and define the computing requirements appropriate to its solution.
c. Design, implement and evaluate computer-based systems or applications to meet desired needs and requirements.
d. Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.
e. Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology
f. Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear
g. Analyze the local and global impact of computing information technology on individuals, organizations and society.

h. Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development.
i. Apply knowledge through the use of current techniques and tools necessary for the IT profession
j. Ability to use and apply current technical concepts and practices in the core information technologies; human computer interaction, information management, programming, networking and web systems and technologies.
k. Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.
l. Integrate IT-based solutions into the user environment effectively.
m. Understand best practices and standards and their applications.
n. Assist in the creation of an effective IT project plan.
o. Ability to demonstrate understanding and proficiency of IT specialization

GRADUATE ATTRIBUTES

These are the skills and qualities our computing graduates will have when they finish the program.

1. Computer Forensics Investigation Process
2. Network Forensics
3. Data Acquisition
4. Malware Forensics
5. Web & Dark Web Forensics
6. Linux & Mobile Forensics
7. Incident Response
8. Evidence Preservation
9. Data Recovery
10. Cyber Law Awareness

COURSE OUTCOMES AND RELATIONSHIP TO PROGRAM OUTCOMES

COURSE LEARNING OUTCOMES (CLO)	**INTENDED LEARNING OUTCOME**	PROGRAM OUTCOMES														
		a	**b**	**c**	**d**	**e**	**f**	**g**	**h**	**i**	**j**	**k**	**l**	**m**	**n**	**o**
1. Understand the Principles of Digital Forensics: Demonstrate comprehension of the foundational principles and methodologies that underpin the field of digital forensics, including its role in cybersecurity investigations.	ILO 1,2		E													
2. Apply Forensic Investigation Processes: Apply systematic and methodical approaches to conduct digital forensic investigations, encompassing evidence identification, preservation, analysis, and presentation.	ILO 3,4,5,6,7					E										
3. Navigate Legal and Ethical Considerations: Navigate the legal and ethical considerations inherent in digital forensics investigations, including privacy laws, evidence admissibility, jurisdictional issues, and adherence to professional ethics standards.	ILO 3,4,5,6,7							E								

WEEK / NO. OF HRS	INTENDED LEARNING OUTCOME (ILO)	DETAILED COURSE CONTENT	TEACHING AND LEARNING ACTIVITY	ASSESSMENT TASK (AT)	CLO
	COURSE ORIENTATION				
1 / 4.83Hours	• Familiarize the student to the course outline • Conduct class regulation and policies	**Orientation of the Course** • Introduction of students and professor • Discussion of the Syllabus • Discussion of Student Outcomes and Program Educational Objectives • Course Requirements • Classroom Policies	Class Discussion • Syllabus • Program Educational Objective • Student Outcomes • Course Requirements • Classroom Policies		
	MODULE 1: Networking Today Basic Switch and End Device Configuration				
1 / 4.83 Hours	• Explain the advances in modern technologies. • Explain how host and network devices are used. • Explain network representations and how they are used in network topologies. • Compare the characteristics of common types of networks. • Explain how LANs and WANs interconnect to the internet. • Describe the four basic requirements of a reliable network. • Explain how trends such as BYOD, online collaboration, video, and cloud computing are changing the way we interact. • Identify some basic security threats and solution for all networks. • Explain employment opportunities in the networking field. • Explain how to access a Cisco IOS device for configuration purposes. • Explain how to navigate Cisco IOS to configure network devices. • Describe the command structure of Cisco IOS software.	**SUBTOPIC 1:** Networking Today -Networks Affect our Lives -Network Components -Network Representations and Topologies -Common Types of Networks -Internet Connections -Reliable Networks -Network Trends -Network Security -The IT Professional •	*Synchronous Activity:* Class Discussion	Formative Assessment	1

	• Configure a Cisco IOS device using CLI. • Use IOS commands to save the running configuration. • Explain how devices communicate across network media. • Configure a host device with an IP address. • Verify connectivity between two end devices. •				
	• Explain how to access a Cisco IOS device for configuration purposes. • Explain how to navigate Cisco IOS to configure network devices. • Describe the command structure of Cisco IOS software. • Configure a Cisco IOS device using CLI. • Use IOS commands to save the running configuration. • Explain how devices communicate across network media. • Configure a host device with an IP address. • Verify connectivity between two end devices.	**SUBTOPIC 2:** Basic Switch and End Device Configuration -Cisco IOS Access -IOS Navigation -The Command Structure -Basic Device Configuration -Save Configurations -Ports and Addresses -Configure IP Addressing -Verify Connectivity	*Asynchronous Activity:* View Video Courseware View Supplementary Materials Consultation with Faculty Members (on top of the assigned consultation) View Guide	Formative Assessment	1
2 / 4.83 Hours	**MODULE 2: Protocols and Models and Physical Layer**				
	• Describe the types of rules that are necessary to successfully communicate. • Explain why protocols are necessary in network communication. • Explain the purpose of adhering to a protocol suite. • Explain the role of standards organizations in establishing protocols for network interoperability. • Explain how the TCP/IP model and the	**SUBTOPIC 1:** Protocols and Models -The Rules -Protocols -Protocol Suites -Standards Organizations -Reference Models -Data Encapsulation -Data Access	*Synchronous Activity:* Class Discussion Guided Formative	Formative Assessment	1

	OSI model are used to facilitate standardization in the communication process. • Explain how data encapsulation allows data to be transported across the network. • Explain how local hosts access local resources on a network.				
	• Describe the purpose and functions of the physical layer in the network. • Describe characteristics of the physical layer. • Identify the basic characteristics of copper cabling. • Explain how UTP cable is used in Ethernet networks. • Describe fiber optic cabling and its main advantages over other media. • Connect devices using wired and wireless media.	**SUBTOPIC 2:** Physical Layer -Purpose of the Physical Layer -Physical Layer Characteristics -Copper Cabling -UTP Cabling -Fiber-Optic Cabling -Wireless Media	*Asynchronous Activity:* View Video Courseware View Supplementary Materials Consultation with Faculty Members (on top of the assigned consultation) View Guide	Formative Assessment	1
3 / 3.67 Hours	**LONG EXAMINATION 1**			Long Examination / Summative Assessment	
3 / 2.83 Hours	**MODULE 3: Number Systems Data Link Layer**				
	• Calculate numbers between decimal and binary systems. • Calculate numbers between decimal and hexadecimal systems.	**SUBTOPIC 1:** Number Systems -Binary Number System -Hexadecimal Number System	*Synchronous Activity:* Class Discussion Guided Formative •	Formative Assessment	1
	• Describe the purpose and function of the data link layer in preparing communication for transmission on specific media. • Compare the characteristics of media access control methods on	**SUBTOPIC 2:** Data Link Layer -Purpose of the Data Link Layer -Topologies -Data Link Frame •	*Asynchronous Activity:* View Video Courseware View Supplementary Materials Consultation with Faculty Members (on top of the assigned consultation) View Guide	Formative Assessment	1

	WAN and LAN topologies. • Describe the characteristics and functions of the data link frame. •				
			• Pre-Summative Assessment Review /Activity (Study Guide and Online Conferencing (Tutorial) • Module 3 Evaluation thru Summative Assessment	Summative Assessment	1
	MODULE 4: Ethernet Switching Network Layer				
4 / 4.83 Hours	• Explain how the Ethernet sublayers are related to the frame fields. • Describe the Ethernet MAC address. • Explain how a switch builds its MAC address table and forwards frames • Describe switch forwarding methods and port settings available on Layer 2 switch ports •	**SUBTOPIC 1:** Ethernet Switching -Ethernet Frame -Ethernet MAC Address -The MAC Address Table -Switch Speeds and Forwarding Methods •	<u>*Synchronous Activity:*</u> Class Discussion •	Formative Assessment	1
	• Explain how the network layer uses IP protocols for reliable communications. • Explain the role of the major header fields in the IPv4 packet • Explain the role of the major header fields in the IPv6 packet • Explain how network devices use routing tables to direct packets to a destination network • Explain the function of fields in the routing table of a router	**SUBTOPIC 2:** Network Layer -Network Layer Characteristics -IPv4 Packet -IPv6 Packet -How a Host Routes -Router Routing Tables	<u>*Asynchronous Activity:*</u> View Video Courseware View Supplementary Materials Consultation with Faculty Members (on top of the assigned consultation) View Guide •	Formative Assessment	1

			• Pre-Summative Assessment Review /Activity (Study Guide and Online Conferencing (Tutorial) • Module 4 Evaluation thru Summative Assessment	Summative Assessment	1
6 / 3.67 Hours	**LONG EXAMINATION 2**			Long Examination / Summative Assessment	
7 / 3.67 Hours	**Midterm Examination**				
8 / 4.83 Hours	**MODULE 5: IPv4 Addressing**				
	• Describe the structure of an IPv4 address including the network portion, the host portion, and the subnet mask • Compare the characteristics and uses of the unicast, broadcast and multicast IPv4 addresses • Explain public, private, and reserved IPv4 addresses	**SUBTOPIC 1:** IPv4 Address IPv4 Address -IPv4 Address Structure -IPv4 Unicast, Broadcast, and Multicast -Types of IPv4 Addresses -Network Segmentation	*Synchronous Activity:* Class Discussion •	Formative Assessment	2
	• Explain how subnetting segments a network to enable better communication • Calculate IPv4 subnets for a /24 prefix • Calculate IPv4 subnets for a /16 and /8 prefix • Given a set of requirements for subnetting, implement an IPv4 addressing scheme •	**SUBTOPIC 2:** IPv4 Subnetting -Subnet an IPv4 Network -Subnet a /16 and a /8 Prefix -Subnet To Meet Requirements •	*Asynchronous Activity:* View Video Courseware View Supplementary Materials Consultation with Faculty Members (on top of the assigned consultation) View Guide	Formative Assessment	1
9 / 4.83 Hours	**MODULE 6: IPv6 Addressing ICMP**				
	• Explain the need for IPv6 addressing • Explain how IPv6 addresses are represented • Compare types of IPv6 network addresses • Explain how to Configure static global unicast and link-local IPv6 network addresses	**SUBTOPIC 1:** IPv6 Address -IPv4 Issues -IPv6 Address Representation -IPv6 Address Types -GUA and LLA Static Configuration -Dynamic Addressing for IPv6 GUAs -Dynamic Addressing for IPv6 LLAs -IPv6 Multicast Addresses -Subnet an IPv6 Network	*Synchronous Activity:* Class Discussion Guided Formative	Formative Assessment	2

<table>
<tr><td></td><td>Explain how to configure global unicast addresses dynamically
Configure link-local addresses dynamically
Identify IPv6 addresses</td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td>Implement a subnetted IPv6 addressing scheme
Explain how ICMP is used to test network connectivity
Use ping and traceroute utilities to test network connectivity</td><td>SUBTOPIC 2: IPv6 Subnetting

-Subnet an IPv6 Network

SUBTOPIC 3: ICMP
- ICMP Messages
- Ping and Traceroute Testing</td><td>Asynchronous Activity:
View Video Courseware

View Supplementary Materials

Consultation with Faculty Members (on top of the assigned consultation)

View Guide</td><td>Formative Assessment</td><td>1</td></tr>
<tr><td></td><td></td><td></td><td>Pre-Summative Assessment Review /Activity (Study Guide and Online Conferencing (Tutorial)
Module 6 Evaluation thru Summative Assessment</td><td>Summative Assessment</td><td>1, 2</td></tr>
<tr><td>10 / 3.67 Hours</td><td colspan="3">LONG EXAMINATION 3</td><td>Long Examination / Summative Assessment</td><td></td></tr>
<tr><td rowspan="2">10 / 4.83 Hours</td><td colspan="5">MODULE 7: Transport Layer Application Layer</td></tr>
<tr><td>Explain the purpose of the transport layer in managing the transportation of data in end-to-end communication
Explain characteristics of TCP
Explain characteristics of UDP
Explain how TCP and UDP use port numbers
Explain how TCP session establishment and termination processes facilitate reliable communication
Explain how TCP protocol data units are transmitted and acknowledged to guarantee delivery
Compare the operations of transport layer</td><td>SUBTOPIC 1: Transport Layer

-Transportation of Data
-TCP Overview
-UDP Overview
-Port Numbers
-TCP Communication Process
-Reliability and Flow Control
-UDP Communication</td><td>Synchronous Activity:
Class Discussion</td><td>Formative Assessment</td><td>1</td></tr>
</table>

	protocols in supporting end-to-end communication •				
	• Explain how the functions of the application layer, presentation layer, and session layer work together to provide network services to end user applications • Explain how end user applications operate in a peer-to-peer network • Explain how web and email protocols operate • Explain how DNS and DHCP operate • Explain how file transfer protocols operate •	**SUBTOPIC 2:** Function of the Application Layer -Application, Presentation, and Session -Peer-to-Peer **SUBTOPIC 3:** Application Layer protocols and Services -Web and Email Protocols -IP Addressing Services -File Sharing Services	*Asynchronous Activity:* View Video Courseware View Supplementary Materials Consultation with Faculty Members (on top of the assigned consultation) View Guide	Formative Assessment	3
11 / 4.83 Hours	**MODULE 8: Network Security Fundamentals Building and Securing a Small Network**				
	• Explain how end user applications operate in a peer-to-peer network • Explain how web and email protocols operate • Explain how DNS and DHCP operate • Explain how file transfer protocols operate	**SUBTOPIC 1:** Network Security Fundamentals -Security Threats and Vulnerabilities -Network Attacks -Network Attack Mitigation -Device Security	*Synchronous Activity:* Class Discussion Guided Formative •	Formative Assessment	1

	• Identify the devices used in a small network • Identify the protocols and applications used in a small network •	**SUBTOPIC 2:** Build a Small Network -Devices in a Small Network -Small Network Applications and Protocols -Scale to Larger Networks -Verify Connectivity -Host and IOS Commands -Troubleshooting Methodologies -Troubleshooting Scenarios	*Asynchronous Activity:* View Video Courseware View Supplementary Materials Consultation with Faculty Members (on top of the assigned consultation) View Guide •	Formative Assessment	1
			• Pre-Summative Assessment Review /Activity (Study Guide and Online Conferencing (Tutorial) • Module 8 Evaluation thru Summative Assessment	Summative Assessment	1
13 / 3.67 Hours	**LONG EXAMINATION 4**			Long Examination / Summative Assessment	
14 / (2 Hours)	**Final Exam**				
TOTAL HOURS	**67.62 hrs**				

CLO	SUMMATIVE ASSESSMENT
1	Summative Assessments 1
2	Summative Assessments 2
3	Summative Assessments 4

GRADINGSYSTEM
FINAL GRADE = 60% CP + 15% Midterm Examination + 25% Final Exam

INSTRUCTIONAL MATERIALS
- Lecture Materials (PPT)/ Supplementary and Reinforcement Materials - Video Materials - Study Guide

TEXTBOOK/REFERENCES/ONLINE REFERENCES

Textbooks:

- Sequeira A. (2017). *CCNA Routing and Switching 200-125 Exam Cram (5th Edition)*
- Cisco Networking Academy (2017). *Introduction to Networks v6 Companion Guide 1st Edition*
- Cisco CCNA In 60 Days (2017). By Paul W Browning (Author), Farai Tafa (Author), Daniel Gheorghe (Author), Dario Barinic (Author) 3rd Edition Createspace Independent Publishing Platform 2017
- Connecting Networks V6 Companion Guide Cisco Networking Academy 1st Cisco Press 2017

Online Sources:

- Courseware Materials available at http://feu.instructure.com
- FEUTECH e-book library collection at https://drive.google.com/drive/folders/15eccuUWwMrcnbHSNMvqn6HWMMyQB3d2?usp=sharing

Journals:

MIS Quarterly
Security Management
Tech & Learning
EBSCO
Journal of Educational Multimedia & Hypermedia

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